Quantifying the Role of Time: A Comparative Study of Real-World Temporal Networks

Pengzhen Deng* Tianqi Hu* Zhangzhang Jiang* Linhua Shen* Keshu Yuan* Ling Feng
Faculty of Science, National University of Singapore
*Equal contribution

Abstract

Real networks consist of interactions that occur at specific times, yet they are routinely analysed as static aggregated graphs. This study quantifies the information that aggregation discards. The temporal-network literature is organised around four questions (representation, activity patterns, structural evolution and function), from which seven testable hypotheses (H1–H7) are derived. A common pipeline of window analysis, randomized reference models and dynamic-community tracking is applied to three networks of different type and time scale: face-to-face contacts in a primary school (PS), online messaging among students (CM) and a computer-science citation network (CIT). The networks are compared through hypothesis-specific time effects rather than raw metrics. The citation network (19,150 papers, 73,269 citations to strictly older papers) shows sublinear degree preference, age-dependent citation choice and densification with exponent 1.343, but no detected shrinkage of the sampled effective diameter; a three-parameter citation model fits selected histories but shows no established advantage over linear extrapolation. In PS and CM, structural metrics depend on the window width and time ordering reduces finite-horizon reachability. The excess burstiness of PS depends on the observation interval and the event definition, and the tie-persistence contrast of CM changes sign with the observation interval and window width. Conditional on major outbreaks, SIR peaks occur later in PS than under both nulls, whereas CM peaks earlier than under timestamp shuffling and close to endpoint randomization. These hypothesis-specific effects do not support a single ranking of network types.

1. Introduction

Real-world networks are dynamic. Nodes and links appear and disappear, and interactions such as face-to-face contacts, messages and citations occur at specific times. Classical network analysis nevertheless aggregates these interactions into a single static graph in which two nodes are either linked or not. Aggregation discards when links are active, how often they recur, in which order they occur and how the network changes. Because connectedness is not transitive in time, a temporal network cannot be reduced to a simple graph without loss of information [12], and the temporal-network literature [12, 13, 22] has shown that each of these features can alter conclusions about structure, communities and dynamical processes such as epidemic spreading.

Most of this evidence derives from a single dataset or a single method, so the extent to which temporal information matters, and for which kinds of system, remains unclear. This study addresses that gap with two aims. The first is to quantify how temporal information changes the characterization of the structure and function of real networks and how this effect differs across network types (RQ1). The second is to determine in which systems the hypotheses H1–H7 drawn from the literature (Tab. 2) hold (RQ2).

Approach. The study is organised around four aspects of temporal networks (Fig. 1). The first, representation, concerns the information lost by aggregation and by the choice of time window. The second, activity patterns, concerns whether interactions are bursty, repeated and persistent. The third, structural evolution, covers growth, ageing and the reorganisation of communities, and the fourth, function, covers the effect of time ordering on reachability, centrality and spreading. Each aspect is translated into hypotheses from the literature, which are operationalised by metrics and tested against randomized reference models [7]. The resulting pipeline is applied to three networks that differ in edge semantics, generating mechanism and temporal resolution: a face-to-face contact network, an online messaging network and a citation network. Because raw metrics are not comparable across such systems, the networks are compared through hypothesis-specific time effects (Sec. 4.7).

Contributions. The study does not introduce new models. Its contribution is a question-driven synthesis of the

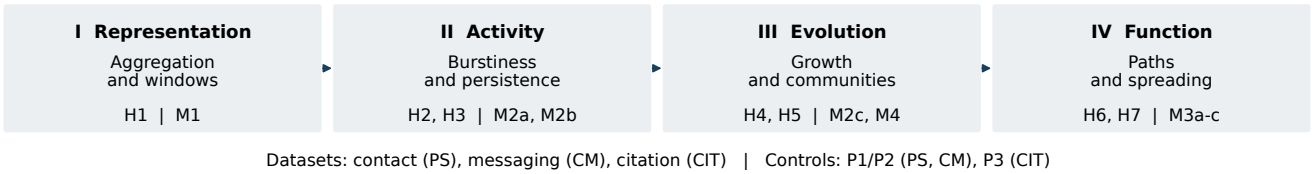


Figure 1. Study design. Four aspects (Parts I–IV) map to seven hypotheses and four method groups, in the order used throughout the paper. Reference models P1/P2 apply to PS and CM, and P3 to CIT.

methods used to study temporal networks, a common and reproducible pipeline that applies these methods to three types of empirical data, and a cross-network assessment of H1–H7 that identifies where temporal information is essential and where a static representation suffices.

Organisation. Section 2 reviews related work, Sec. 3 describes the data and Sec. 4 the methods. Section 5 reports the results, Sec. 6 discusses them and Sec. 7 concludes.

2. Related Work

The literature is organised by question, following Parts I–IV of the Introduction. General frameworks and reviews are given by Holme [12], Holme and Saramäki [13] and the book of Masuda and Lambiotte [22]. A central point of these reviews is that the basic unit is a *contact* (two nodes and a time), and that connectedness is no longer transitive: if a meets b and later b meets c, a path from a to c exists, but not the reverse path. Consequently, a temporal network cannot be mapped onto a simple graph without losing information or redefining its nodes, and many static-network methods require non-trivial generalisations [12].

2.1. Representing time: aggregation, snapshots and windows

Temporal networks can be stored as contact sequences, interval graphs or sequences of snapshots [13]. Lossy representations such as weighted aggregates or reachability graphs reduce the data to a static object; lossless ones keep all contacts [12]. Each choice involves a trade-off: snapshots make static tools applicable but require a window length, whereas the full event sequence retains all information. The effect of this choice has been quantified for communication data: window size and placement change the structure of the aggregated network [18], and coarser temporal resolution distorts properties of the time-varying network in a way that depends on the system [31]. These findings motivate **H1**: aggregation and window choice change structural conclusions.

2.2. Temporal activity patterns

Human activity is bursty: inter-event times are heavy-tailed rather than exponential [1], and consecutive intervals are correlated, which Goh and Barabási [9] capture with a burstiness coefficient and a memory coefficient. Social ties are also repeated and persistent over long periods [34], while the set of active links turns over. Temporal motifs [17, 28] and activity-driven models [29] describe and generate such patterns. These observations motivate **H2** (activity is bursty and has memory) and **H3** (ties are repeated and persistent while the network turns over).

2.3. Structural evolution: growth, ageing and communities

Growth is classically explained by preferential attachment [2], whose strength can be measured directly from growth data [14, 25]. Real systems usually show ageing, where old nodes attract fewer new links [5], and in citations this is captured by a relevance decay [38]. Growing networks also densify, with the number of edges growing superlinearly in the number of nodes [19, 20]; node and link deletion alter these laws [23]. These results motivate **H4**: growth follows preferential attachment with ageing, and networks densify.

At the mesoscale, multilayer modularity couples partitions across time [24], communities can be tracked across snapshots through matching [10] and change can be mapped between partitions [33]; Rossetti and Cazabet [32] survey the field. Per-snapshot partitions are typically obtained with Louvain [3] or Leiden [37]. These methods motivate **H5**: communities reorganise over time and a static partition hides this.

2.4. Time and network function

Because interactions occur in sequence, only time-respecting paths can carry information: reachability in contact sequences is much lower than in the aggregated graph [11], and path lengths and centrality change accordingly [16, 26]. Burstiness and temporal correlations further slow down spreading even when the static topology is unchanged [15]. These results motivate **H6** (time ordering restricts reachability and changes centrality rankings) and **H7** (burstiness and temporal correlations slow down spreading).

Type	Dataset	N	Events	Span	Res.
Contact	Primary school (PS)	242	125 773	2 d	20 s
Comm.	CollegeMsg (CM)	1 899	59 798	193 d	1 s
Citation	DBLP–DB (CIT)	19 150	73 269	50 yr	1 yr

Table 1. Datasets. N: nodes; Res.: temporal resolution. CIT spans publication years 1970–2020; the primary analysis covers 1990–2020. PS is undirected and has class labels; CM is directed and unlabelled; CIT is directed and acyclic.

2.5. Randomized reference models

Randomization is more versatile for temporal than for static networks, for which it is essentially limited to degree-preserving link shuffling, because individual types of correlation can be removed selectively [12]. Randomized reference models act as hypothesis tests: each one preserves some properties of the data and destroys others, so differences between data and model reveal which feature matters [7]. Such models are used in Parts II–IV (Sec. 4.6).

Time scales and network types. Temporal networks are usually defined as systems whose evolution is not much slower than the processes running on them, and a single system contains many time scales (seconds, hours, days) [12]. Contact and communication data are event sequences with such a hierarchy, whereas citation networks are acyclic by construction and all out-links of a paper appear at one instant [12]. Different network types are therefore expected to differ in their sensitivity to temporal information.

Research gap. Most studies examine one dataset or one method, and few compare within a single framework how sensitive different *types* of networks are to temporal information. The hypotheses distilled above (Tab. 2) enable such a comparison.

3. Data

One network is used per type. The three networks were chosen to differ in edge semantics, generating mechanism and temporal resolution; their properties are summarised in Tab. 1.

Face-to-face contacts. The primary-school data [8, 35] were collected by the SocioPatterns collaboration with wearable proximity sensors over two school days. A contact is recorded when two individuals are face to face within approximately 1.5 m for at least 20 s. Each individual carries a class label, which serves as ground truth for community detection. The data are released under CC BY-NC-SA 3.0. Activity is confined to school hours (approximately 08:30–17:20), with up to 2,168 contact records per 10 min and no records overnight.

Online communication. CollegeMsg [27] contains time-stamped private messages between 1 899 users of an online student community over 193 days at 1 s resolution; removing 37 duplicate triples leaves 59,798 messages. Edges are directed, and no node labels are available. Activity is strongly nonstationary: 37,680 messages (63%) fall in May 2004, and monthly volume declines steadily thereafter to 816 messages in October.

Scientific publications. The citation network is an induced subgraph of AMiner DBLP-Citation V14 [36], restricted to a fixed whitelist of database venues (SIGMOD, VLDB/PVLDB, ICDE, PODS, EDBT, ICDT, TODS, VLDB Journal, Data & Knowledge Engineering and Information Systems) and their declared name variants. Of 5,259,858 streamed records, 19,150 publications from 1970–2020 are retained, including 6,709 isolates. A citation $u \to v$ is dated by the publication year of u, and the primary graph requires $y_v < y_u$; it contains 73,269 distinct citations and is therefore acyclic. The 5,486 same-year references, whose order cannot be resolved, are kept separately for a sensitivity analysis; 6,390 references to later papers, 494 self-loops and 37,440 repeated reference entries are removed. Author self-citations are retained. All counts refer to citations within the selected venues and not to global citation impact. AMiner permits research use; no further licence is assumed. The number of retained citations per year declines in the most recent years although the number of publications continues to grow, which indicates coverage and censoring effects and motivates a sensitivity check with a 2015 cutoff.

Preprocessing. All data are converted to a common event format (u, v, t) with integer timestamps (seconds or years) and a table of node labels. Self-loops and exact duplicate events are removed. Each source has known limitations: the citation data have annual resolution, the contact data cover only two days, and the sensors record only participating individuals. The datasets differ by design in edge semantics (proximity, message, citation), directedness and time scale, so that agreement across them cannot be attributed to a single data type.

4. Methods

Every analysis follows the same template: a hypothesis from the literature is operationalised by a metric and tested against a randomized reference model. Table 2 lists the seven hypotheses, the methods (M1–M4) that test them and the datasets for which each method is meaningful (19

method–dataset pairs). Some combinations are inapplicable by construction: CIT is a directed acyclic graph with annual resolution, without fine-grained events (M2a, M2b, M3) or community reorganisation (M4), whereas PS and CM have no growth process (M2c).

4.1. Notation

Let $\mathcal{E} = \{(u, v, t)\}$ be the event sequence. The static aggregate G contains a link (u, v) if at least one event occurs between u and v. A snapshot $G_{\Delta t}(k)$ contains the events in $[k\Delta t, (k + 1)\Delta t)$, and the cumulative network $G(\leq t)$ contains all events up to t.

4.2. Aggregation and time windows (M1)

The observation period is divided into non-overlapping, left-closed windows of width Δt, starting at the first event and ending one timestamp resolution after the last event. Widths are 5, 15 and 30 min and 1, 2, 4, 8 and 24 h for PS, and 1, 3, 6 and 12 h and 1, 3, 7 and 30 days for CM. Each distinct pair with at least one event in a window forms one unweighted edge, undirected in PS and directed from sender to recipient in CM. Every snapshot contains the full node set of the dataset (242 and 1 899 nodes), including isolates, and empty windows are retained, so that overnight inactivity in PS enters the averages. For each snapshot, density, mean total degree, mean local clustering (including zeros), the number of components and the fraction of nodes in the largest component (LCC) are computed. CM uses in- plus out-degree, directed clustering [6] and weakly connected components. Metrics are averaged with equal weight over the complete windows of each width; the shorter final window is evaluated separately as a boundary check. The *window sensitivity* of a metric m is $(\max m - \min m)/|\overline{m}|$ over the window-averaged values on the grid. Under a fixed node set, density and mean total degree are proportional and therefore constitute a single effect. Halves of the observation period are compared on the widths that have complete windows in both halves, which excludes the 24-h width for PS. For CIT, annual and cumulative networks are compared, together with trailing windows (Sec. 4.4). Aggregation can also introduce paths that cannot be traversed in time: if c–d occurs before a–b and b–c, the static path a–b–c–d is not time-respecting from a; this effect is measured in M3.

4.3. Activity patterns (M2a, M2b)

M2a: burstiness and memory. Inter-event times are computed within each interacting node pair. Because PS contacts are sampled every 20 s, consecutive records of the same unordered pair 20 s apart are merged into a single contact episode; in CM each directed message is an independent event. Intervals are pooled across pairs, and burstiness is

$$B = \frac{\sigma - \mu}{\sigma + \mu}, \tag{1}$$

where μ and σ are the mean and population standard deviation of the pooled intervals. Memory M is the Pearson correlation between consecutive intervals within a pair, pooled over all adjacent interval pairs and never across different pairs [9]. Both B and M are pooled statistics and not averages of per-pair coefficients. The shape of the pooled interval distribution is assessed by comparing exponential and lognormal maximum-likelihood fits by AIC.

M2b: repetition, lifetime and turnover. The repetition rate is the fraction of distinct pairs with at least two contact episodes (PS) or two messages (CM). The lifetime of a pair is the time between its first and last event, so that pairs observed once have zero lifetime. Turnover is measured by the Jaccard similarity of the edge sets of consecutive windows,

$$J(E_k, E_{k+1}) = \frac{|E_k \cap E_{k+1}|}{|E_k \cup E_{k+1}|}, \tag{2}$$

averaged over adjacent window pairs with a nonempty union. The default window width is 1 h for PS and 7 days for CM; widths of 15–120 min (PS) and 1–14 days (CM) are used as sensitivity checks. Lifetimes and Jaccard similarities are computed from the original records, and the comparison of raw records with merged episodes assesses the sensitivity of H2 to the event definition.

4.4. Structural evolution (M2c, M4)

M1 for citations. Annual and cumulative citation networks share the same node set at year t, namely all selected papers published by t. Directed density is $E/[N(N-1)]$ and mean total degree is $2E/N$; clustering, component counts and LCC fractions use the weak undirected projection, including isolates. Trailing windows of 1, 2, 5 and 10 years provide the window grid. Window sensitivity, $(\max - \min)/\mathrm{mean}$ over the grid, is computed at each end year and averaged over end years 1990–2020, which constitute the primary analysis interval; earlier papers enter as history.

M2c-1: preferential attachment. Following the degree-class approach of Jeong et al. [14], in-degrees are frozen at the start of each citing year, and for each prior-degree class k the number of eligible papers n_{tk} and of new citations h_{tk} are counted. The exponent α is estimated by conditional likelihood with target probabilities proportional to $n_{tk}(k+1)^{\alpha}$, conditioning on the total number of citations in each year. The offset admits citations to previously uncited papers; offsets of 0.5 and 2 serve as sensitivity checks. Confidence intervals use 400 bootstrap resamples of citing-year blocks and are descriptive. Neither a positive α nor its departure from an age-only reference identifies a causal mechanism.

Part	ID	Hypothesis	Source	Method	Null	Time effect	PS	CM	CIT
I	H1	Aggregation and window choice change structural conclusions	[18, 31]	M1	–	Window sensitivity	✓	✓	✓
II	H2	Activity is bursty and has memory	[9]	M2a	P1	ΔB, ΔM	✓	✓	
	H3	Ties are repeated and persistent, and the network turns over	[34]	M2b	P1	ΔJ	✓	✓	
III	H4	Growth follows preferential attachment with ageing, and networks densify	[14, 19, 38]	M2c-1–4	P3	$\Delta\alpha$, a			✓
	H5	Communities reorganise over time, which the static partition hides	[10, 24]	M4	P1	Δ NMI	✓	✓	
IV	H6	Time ordering restricts reachability and changes centrality rankings	[11, 16, 26]	M3a, M3b	P1, P2	ΔR, τ	✓	✓	
	H7	Burstiness and temporal correlations slow down spreading	[15]	M3c	P1, P2	$T_{\mathrm{real}}/\overline{T}_{\mathrm{null}}$	✓	✓	

Table 2. Hypotheses, sources, methods, reference models (Sec. 4.6), time-effect measures (Sec. 4.7) and applicable datasets.

M2c-2: ageing. For ages 1–60, the target probability in citing year t is proportional to $n_{ta}f(a)$, where n_{ta} is the number of available papers of age a. Lognormal, exponential, power-law and uniform kernels f are compared by AIC on identical observations and support. The comparison is repeated by publication cohort (1970–1989, 1990–1999, 2000–2009), accounting for each cohort's exposure. The estimate describes marginal ageing, which may also reflect degree, fitness and venue coverage.

M2c-3: densification and diameter. The densification exponent a is obtained from $\log E(t) = a \log N(t) + b$ over 1990–2020 and over split periods and an earlier cutoff. Year-resampling intervals are descriptive because cumulative observations are dependent. The effective diameter is the 90th percentile of shortest-path distances within the largest weak component, estimated from 128 uniformly sampled source nodes per year, with 64 and 256 sources as sensitivity checks [19].

M2c-4: citation dynamics and prediction. The Wang–Song–Barabási (WSB) model [38],

$$c(t) = m\left[\exp\left\{\lambda\Phi\left(\frac{\log t-\mu}{\sigma}\right)\right\}-1\right], \qquad (3)$$

is fitted with m fixed to the mean within-subgraph reference count of papers published up to 1989. Bounded multistart least squares minimizes the squared error of $\log(1+c)$; parameter bounds and boundary flags are recorded. Descriptive fits use a random sample of at most 500 papers published in 1990–2005 with at least 10 citations by age 15, fitted over ages 1–15; the resulting collapse is in-sample evidence only. For prediction, papers are selected using only their first five years (at least five citations by age five; random sample of at most 500), and WSB is compared with a no-ageing preferential attachment baseline, $c(t) = m[\exp(rt)-1]$, and with linear extrapolation, $c(5)t/5$. The primary task predicts cumulative citations at age 15, which is observed for every selected paper; a secondary task predicts the 2020 total with a variable horizon. Neither target represents lifetime citations. This fixed-horizon design differs from the original study, which trains on 5 or 10 years and evaluates at age 30 [38]. Performance is measured by mean absolute error (MAE), median absolute error and root mean squared logarithmic error (RMSLE), with paired bootstrap differences (1000 resamples). Fits at a parameter bound remain in the primary evaluation, and an interior-fit subset serves as a sensitivity check; a constrained optimum is not regarded as a convergence failure.

P3: citation-target randomization. Twenty realizations per model preserve each paper's publication year and its number of distinct references, which are drawn without replacement from strictly older papers. P3a selects targets uniformly. P3b weights targets by a nonparametric age kernel fitted by conditional likelihood to the observed data and removes degree preference. Because P3b is calibrated on the data, its agreement with the observed ageing is not independent confirmation. Null means and sample standard deviations are reported; empirical tail probabilities have a resolution of $1/21$. Both models fix $N(t)$ and $E(t)$ and thus preserve the densification exponent exactly; they test diameter and target preference but not densification itself.

M4 cuts the event sequence into consecutive windows of width Δt and aggregates each window into a weighted undirected snapshot whose weights are event counts. Windows with fewer than 100 events (PS; this removes nights) or 50 events (CM) are discarded, and each snapshot contains only the nodes active in its window. The main settings

are $\Delta t = 15\,\text{min}$ for PS and 7 days for CM. Every snapshot and the static aggregate are partitioned with Leiden [37] maximising modularity with resolution γ [30]. A multislice partition [24] with interslice coupling $\omega = 1$ gives labels that are consistent across windows. It is used only for tracking, because the coupling itself draws consecutive partitions together.

Partitions are compared by normalized mutual information (NMI) [4] on the nodes they share. For each window, the NMI is recorded with the previous window (for consecutive windows only, not across a night), with the static partition and, for PS, with the classes (teachers excluded). As a noise floor, every window is partitioned a second time with a different random seed. All measures are recomputed on 20 P1 samples. P1 preserves the static graph, and hence the static partition, as well as the number of events per window; H5 therefore predicts that real snapshots agree less with each other and with the static partition than P1 snapshots do. The analysis is repeated for $\gamma \in \{0.5, 1, 2\}$ and $\Delta t \in \{5, 15, 60\}$ min (PS) or $\{1, 7, 30\}$ days (CM).

4.5. Paths, centrality and spreading (M3)

M3a computes earliest-arrival reachability from 16 clock-hour start times in PS and 20 evenly spaced start times within the first 90% of the CM interval. Paths require strictly increasing timestamps, so simultaneous events cannot form a chain, and unreachable pairs contribute zero. The reachability ratio R divides temporal reachability by the static reachability of the full selected period on the same node set and with the same edge direction; late start times therefore include a finite-horizon effect. Mean latency is averaged over reachable pairs at each start time and then over start times.

M3b compares static and temporal closeness. Static closeness is harmonic closeness [21], $C^{\mathrm{s}}(i) = \sum_{j \neq i} 1/d(i,j)$, on the aggregate (outgoing distances for CM), with hop distance (unweighted) or distance $1/w_{ij}$ (weighted). Temporal closeness [16, 26] uses the same time-respecting paths as M3a: from a start time t_0, an event (u, v, t) can be used only if u was reached strictly before t, and v is then reached at t, so no path takes two hops within one timestamp. With $a_{t_0}(i \to j)$ the earliest arrival time,

$$C^{\mathrm{t}}(i) = \frac{1}{|\mathcal{T}|} \sum_{t_0 \in \mathcal{T}} \frac{1}{N-1} \sum_{j \neq i} \frac{1}{a_{t_0}(i \to j) - t_0}, \quad (4)$$

where unreachable nodes contribute zero. $\mathcal{T}$ holds every full hour from 09:00 to 16:00 on both PS days (16 start times) and 20 evenly spaced times over the first 90% of the CM period. Agreement is measured by Kendall's τ_b and the overlap of the top 10% nodes. P1 leaves the static graph and so C^{s} unchanged; recomputing C^{t} on 20 P1 samples therefore isolates the effect of timing.

M3c simulates an event-driven SIR process from 200 seeds per network. Seed events are drawn uniformly from the first 70% of the event sequence; the source is the sender in CM and a random endpoint in PS. The source is infected at the time of the seed event, and only later events transmit, each with probability β; the infectious duration D is fixed, and recovered nodes remain immune. Four parameter settings are used per network, the first being primary: $(\beta, D) = (0.05, 4\,\text{h}), (0.02, 4\,\text{h}), (0.10, 4\,\text{h}), (0.05, 1\,\text{h})$ for PS and $(0.20, 14\,\text{d}), (0.05, 14\,\text{d}), (0.50, 14\,\text{d}), (0.20, 3\,\text{d})$ for CM. The outcomes are the final infected fraction, the probability of a major outbreak (at least 5% of the nodes infected) and the time T to the prevalence peak. Simultaneous infections and recoveries are combined before prevalence is measured. Peak times are averaged over major outbreaks only, so the ratio $T_{\mathrm{real}}/\overline{T}_{\mathrm{null}}$ is a conditional ratio of means; a network without a major outbreak yields an undefined ratio and not a zero delay. Each run uses 20 P1 and 20 P2 networks.

4.6. Randomized reference models

No single temporal counterpart of the configuration model exists; instead, selected correlations are removed by different models and the outcomes are compared [7, 13]. For PS and CM, two complementary randomizations are used, corresponding to *randomly permuted times* (RP) and *randomized edges* (RE) in the terminology of Holme and Saramäki [13].

P1 (RP): timestamp permutation. Timestamps are permuted among events while the endpoints of each event are kept. P1 preserves the static graph, the number of events of every pair and the global timestamp distribution, and destroys temporal correlations. Coincident events produced by the permutation are retained to preserve event counts. Twenty realizations are used unless stated otherwise.

P2: endpoint swaps. Endpoints are swapped between randomly chosen event pairs while timestamps are kept fixed. In undirected PS, endpoint orientations are randomized before swapping; in directed CM, sender and receiver roles are preserved. Swaps that would create self-loops or duplicate events are rejected, and five swaps are attempted per event. P2 preserves the number of events, the multiset of timestamps and the event participation count of every node (separately as sender and receiver in CM), but not the static degrees or the event sequences of individual pairs. These invariants are verified on a fixed seed; uniform sampling and sufficient mixing are not established. A degree-preserving rewiring that moves each pair's event sequence with its link [13] and combined models (RE+RP) were not evaluated. Comparing P1 with P2 separates the effects of event timing from those of the topology.

P3, the reference model for citations, is described in Sec. 4.4.

4.7. Measuring the effect of time

Raw metrics cannot be compared across a contact network, a message network and a citation network, so each hypothesis is assigned one time-effect measure (Tab. 2). For H2–H6 the effect is the difference $\Delta x = x_{\mathrm{real}} - \overline{x}_{\mathrm{null}}$ between the empirical value and the mean over the stated reference model, expressed in the units of the underlying metric. H1 has no reference model and is summarised by the descriptive window sensitivity, H4 additionally reports the densification exponent a, and H7 is summarised by the ratio $T_{\mathrm{real}}/\overline{T}_{\mathrm{null}}$ of the times to the prevalence peak. A common difference operator does not place different hypotheses on a common scale, and quantities that are invariant under P1 are reported as descriptive values and not as evidence from a null test.

4.8. Robustness protocol

For PS and CM, the observation period is split at the elapsed-time midpoint and all analyses are repeated on both halves. PS start times retain their clock-hour convention, whereas CM start times and SIR seeds are redrawn within each half. The null ensemble is enlarged from 20 to 60 networks, nested over the first 20, with the same empirical network. SIR retains all four parameter settings, and the community analysis repeats its primary window and resolution on the halves. For CIT, the 1990–2020 analysis interval is split into 1990–2004 and 2005–2020, with earlier papers retained as eligible targets; the attachment and ageing statistics receive additional P3 realizations. The fixed-horizon forecast test of M2c-4 is not repeated on the halves.

4.9. Implementation

All code is written in Python. NetworkX is used for static graphs, a strict-time event scan for earliest arrivals, NumPy with optional Numba acceleration for SIR, and igraph/Leiden for per-snapshot and multislice partitions. The reference models, the attachment-kernel estimator and the SIR process are custom implementations, unit-tested on small examples with known answers (reachability, SIR and the invariants of P1 and P2). Random seeds are fixed, input checksums and all randomized edge lists are stored, and each figure is generated by a single script from saved results. Computationally intensive workloads were run on a CPU cluster.

5. Results

Results follow the four method groups of Tab. 2. Each subsection states the measurement protocol, reports the measured values and closes with a verdict on the corresponding hypotheses. Null comparisons use 20 randomized networks per family. Sensitivity to the observation interval and to the number of null networks is reported in Sec. 5.5, and the cross-network summary in Sec. 5.6.

Metric	PS	CM	CIT
Density (= mean total degree)	3.262	5.411	1.798
Clustering	1.578	5.671	2.361
Components	1.712	0.398	–
LCC fraction	2.096	4.115	1.114

Table 3. Window sensitivity (relative range of complete-window means over the grid: PS 5 min–24 h, CM 1 h–30 days; CIT trailing windows of 1–10 years averaged over end years). The values are descriptive and are not tested against a reference model.

5.1. Representation and time windows (H1)

Contact and messaging networks. Every structural metric changes substantially with the window width in both networks (Tab. 3, Fig. 2a,b). Wider windows combine interactions that occurred at different times and therefore increase density, mean degree, clustering and connectivity while reducing the number of components. In PS the mean LCC fraction rises from 11.6% in 5-min windows to 56.3% in 8-h windows and reaches 99.6% in the single complete 24-h window. In CM it rises from 0.24% at 1 h to 35.8% at 30 days (six complete windows). The number of components in CM varies least (sensitivity 0.398), because most users remain isolated in all but the widest windows.

Citation network. In 2020 the annual and cumulative graphs on the same 19,150 published papers contain 1,450 and 73,269 citations, with mean total degree 0.151 and 7.652 and LCC fraction 0.052 and 0.641, respectively. Cumulative clustering is 0.124, whereas annual clustering is identically zero: because citations point to strictly older papers, every annual snapshot is bipartite. This is a property of the representation and not evidence of temporal correlation. Over trailing windows of 1–10 years, the sensitivities are 1.798 for density, 2.361 for clustering and 1.114 for the LCC fraction (Fig. 2c). Longer windows connect a larger fraction of papers and admit triangles spanning several years, which a single final aggregate cannot reveal.

Observation halves. The same directions hold at every comparable width in both halves of both networks. On the common grid, the PS sensitivities are stable (density 2.503 over the full period against 2.551 and 2.518 in the halves; clustering 1.394 against 1.470 and 1.487). The magnitude of connectivity in CM, however, depends on the period: at 30 days the mean LCC fraction is 53.3% in the first half and 18.6% in the second (three complete windows each), which coincides with a decline in activity from 53,163 to 6,635 messages.

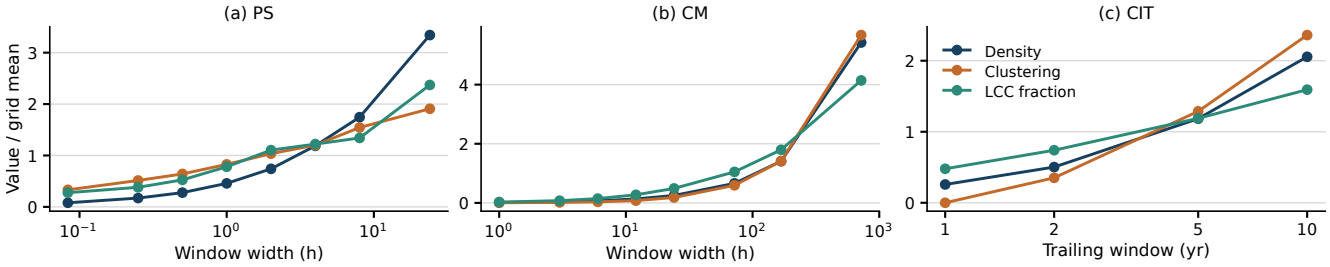


Figure 2. Window dependence of structural metrics. Each metric is divided by its mean over the window grid, so the vertical extent of a curve equals its window sensitivity (Tab. 3). CIT curves are averaged over end years 1990–2020.

Network	Metric	Real	P1 mean ± SD
PS	B	0.535	0.5184 ± 0.0004
PS	M	0.020	0.0710 ± 0.0016
PS	Repeated ties	0.625	0.6568 ± 0.0001
PS	Jaccard	0.220	0.3668 ± 0.0010
CM	B	0.581	0.2451 ± 0.0012
CM	M	0.100	0.2288 ± 0.0057
CM	Repeated ties	0.495	0.4949 ± 0.0000
CM	Jaccard	0.096	0.0690 ± 0.0009

Table 4. Burstiness B, memory M, repeated-tie fraction and adjacent-window Jaccard similarity (1-h windows for PS, 7-day windows for CM). The CM repeated-tie fraction is invariant under P1 and therefore descriptive.

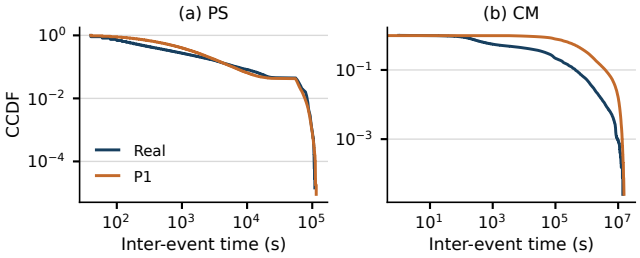


Figure 3. Complementary cumulative distribution of within-pair inter-event times (PS: contact episodes), real network and one P1 realization.

Verdict on H1. H1 is supported in all three networks as a representation effect. The numerical sensitivity depends on the window grid, the window alignment, the inclusion of inactive nodes and the observation span, so the values in Tab. 3 do not rank the network types. Connectivity after aggregation also discards event order; whether aggregated paths can be traversed in time is examined under H6.

5.2. Activity patterns (H2, H3)

Table 4 compares the activity statistics of both networks with P1.

Network	Setting	Real	Δ vs P1
Adjacent-window Jaccard J by window width			
PS	15 min	0.234	−0.023
PS	30 min	0.217	−0.105
PS	1 h	0.220	−0.147
PS	2 h	0.202	−0.144
CM	1 d	0.089	+0.071
CM	3 d	0.100	+0.059
CM	7 d	0.096	+0.027
CM	14 d	0.089	−0.017
PS event definition			
B	Raw 20-s records	0.628	+0.098
B	Contact episodes	0.535	+0.016
M	Raw 20-s records	0.034	−0.041
M	Contact episodes	0.020	−0.051

Table 5. Sensitivity of H2 and H3 over the full period (P1: 20 realizations, SD ≤ 0.0017). The primary settings are 1 h (PS), 7 d (CM) and contact episodes.

Burstiness and memory (H2). Burstiness exceeds the P1 value in both networks, by $\Delta B = +0.016$ in PS and $+0.336$ in CM, whereas memory is lower than under P1 in both ($\Delta M = -0.051$ and -0.129). The PS excess is small in absolute terms over the full interval, although large relative to the narrow null dispersion; it increases to $+0.116$ and $+0.120$ in the two single-day halves (real/null 0.404/0.288 and 0.395/0.275). The inactive night between the two school days is a plausible contributor but cannot be isolated with these data. The mean tie lifetime is 0.444 days against 0.690 days under P1 in PS, and 4.756 days against 30.028 days in CM.

The magnitude of the PS excess also depends on the event definition (Tab. 5). On raw 20-s records, consecutive records of one contact create many short intervals, and the excess rises to $+0.098$; merging them into episodes reduces it to $+0.016$. Memory remains below P1 under both definitions. The inter-event distributions are heavy-tailed (Fig. 3): a lognormal fit is preferred over an exponential one by $\Delta\mathrm{AIC} = 1.9 \times 10^5$ (PS) and 1.8×10^5 (CM), with

coefficients of variation of 3.3 and 3.8. In PS, however, the P1 distribution is similarly heavy-tailed (coefficient of variation 3.2), because the school timetable and the night are retained by timestamp permutation; the burstiness of PS is therefore largely a property of the global schedule. In CM, real intervals are much shorter than under P1 (median 41 min against 5.4 days), which is consistent with rapid exchanges within conversations.

Persistence and turnover (H3). Adjacent-window similarity is lower than under P1 in PS ($\Delta J = -0.147$), indicating rapid turnover of the contact set. In CM the full-period similarity exceeds that of P1 ($\Delta J = +0.027$), but the sign reverses within both halves (0.074 against 0.113 and 0.118 against 0.140) and for 14-day windows (-0.017), whereas 1- and 3-day windows give larger positive contrasts (Tab. 5). In PS the contrast is negative at every width and largest from 30 min upwards. The CM contrast thus depends on both the observation interval and the window width: P1 redistributes timestamps over the selected interval, so its baseline varies with the duration, activity level and window composition of that interval, and the present comparison does not identify which of these factors is responsible. The repeated-tie fraction of CM is invariant under P1 by construction and serves only as a description.

Verdict on H2 and H3. H2 is supported for burstiness in CM, where the excess is large and robust. In PS the excess is small and depends on the observation interval and the event definition, and heavy-tailed intervals arise under P1 as well. H2 is not supported for memory, which is weaker than under P1 in both networks. For H3, PS shows rapid turnover at every width, whereas in CM ties persist beyond P1 at short and intermediate widths but not at 14 days or within the halves, so the data do not support a general claim of more persistent ties.

5.3. Structural evolution (H4, H5)

5.3.1. Growth, ageing and densification of the citation network (H4)

Preferential attachment (H4a). The conditional attachment kernel $(k+1)^{\alpha}$ yields $\alpha = 0.714$ with a descriptive 95% year-bootstrap interval $[0.693, 0.730]$ (Fig. 4a). Twenty P3a samples give 0.0006 ± 0.0051 and the age-only control P3b gives -0.6512 ± 0.0053 (Fig. 4b); the observed exponent exceeds all null values, corresponding to an empirical one-sided probability of $1/21$ per comparison. The exponent remains positive under time splits, degree offsets of 0.5 and 2, and a core subset of venues (Sec. 5.5). Sublinear degree preference is therefore supported within this corpus beyond the age-only reference, without identifying a causal attachment mechanism.

Age kernel	Parameters	ΔAIC
Lognormal	2	0.0
Exponential	1	537.8
Power-law	1	9,360.5
Uniform	0	41,796.7

Table 6. CIT age-preference kernels fitted on identical observations. AIC selects among the stated models and not among all possible ageing mechanisms.

Ageing (H4b). A lognormal age kernel is preferred over exponential, power-law and uniform kernels (Tab. 6). The fitted parameters are $(\mu, \sigma) = (1.858, 1.108)$, corresponding to a scale e^{μ} of 6.41 years. The mean age of cited papers is 5.711 years, against 11.457 ± 0.034 under P3a and 5.717 ± 0.018 under the calibrated control P3b (Fig. 4c). The lognormal kernel is preferred in all three publication cohorts, although the tails of the most recent cohort remain censored. Age-independent target choice is therefore rejected; the agreement of P3b follows from its calibration and does not constitute independent confirmation.

Densification and diameter (H4c). For 1990–2020 the densification exponent is $a = 1.343$ ($R^2 = 0.991$) with a descriptive year-resampling interval $[1.288, 1.406]$ (Fig. 4d). Densification is weaker in 2005–2020 ($a = 1.065$) than in 1990–2004 ($a = 1.455$). Here densification denotes superlinear growth of the edge count and not increasing directed density. The effective diameter of the largest weak component equals 6 hops in every year and is unchanged with 64 or 256 source nodes, so no shrinkage is detected at this resolution. Because P3 fixes $N(t)$ and $E(t)$, it preserves a by construction and cannot test densification.

Citation dynamics and prediction (H4d). For 500 full-history fits, the median WSB parameters are $(\lambda, \mu, \sigma) = (2.717, 0.834, 1.238)$, with median RMSLE 0.074 and scaled-collapse RMSE 0.022; these are in-sample diagnostics. In the held-out task (Tab. 7), WSB predicts cumulative citations at age 15 with an MAE of 15.144, against 17.064 for linear extrapolation. The paired difference (WSB minus linear) has a 95% bootstrap interval of $[-4.471, 0.806]$ at age 15 and $[-4.764, 8.128]$ for the 2020 total; both intervals contain zero. The no-ageing baseline over-extrapolates by several orders of magnitude, so outperforming it alone is a weak test. Forecast fits reach a parameter bound for 193 of 500 papers (38.6%), although all optimizations report success; boundary solutions reflect limited identifiability in short histories rather than optimizer failure. Compact descriptive fits are thus supported, whereas a predictive advantage over linear extrapolation is not established.

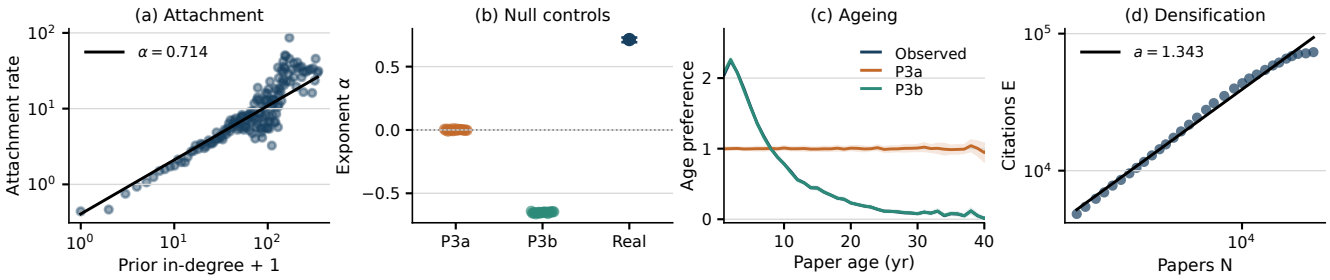


Figure 4. CIT growth mechanisms. (a) Attachment rate by prior-degree class (classes with at least five citations); the line shows the fitted exponent. (b) Exponents of 20 samples per reference model and the observed exponent with its year-bootstrap interval. (c) Exposure-adjusted age preference (null mean ± SD). (d) Cumulative edge count against node count, 1990–2020.

Target	Model	MAE	RMSLE
Age 15	WSB	15.144	0.601
	Linear	17.064	0.662
	No-ageing PA	27,588	4.601
2020	WSB	28.016	0.778
	Linear	27.271	0.876
	No-ageing PA	2.79×10^{6}	7.000

Table 7. Citation forecasts for 500 papers trained on years 1–5. Age 15 is the primary, fixed-horizon target; the 2020 total is a secondary target with variable horizon.

Verdict on H4. Sublinear preferential attachment, age-dependent target choice and superlinear edge growth are supported within this corpus, whereas diameter shrinkage and a forecast advantage of the WSB model are not. All counts are restricted to the selected venues.

5.3.2. Dynamic and static communities (H5)

Primary school. Outside the midday break, snapshot communities closely match the classes (NMI ≈ 0.85–0.95), as does the static partition (six communities, NMI 0.86 with the classes; Fig. 5). Between approximately 12:00 and 14:00 on both days, the NMI with the classes and with the static partition falls to ≈ 0.5. Under P1 both curves remain flat (class NMI 0.86 ± 0.002), so the decline is attributable to the timing of contacts and not to the identity of the contacting pairs. Nor is it algorithmic noise: two Leiden runs on the same window agree with NMI 0.98 on average. Averaged over the day, real snapshots agree less with the static partition than P1 snapshots (0.72 against 0.94 ± 0.003).

CollegeMsg. Weekly snapshots agree only weakly with each other and with the static partition (NMI 0.52 and 0.34), and P1 yields similar values (0.42 and 0.32). Snapshots are sparse, with on average 326 of 1,899 users active per week, and during the first two months two runs of the algorithm on the same window agree only at NMI ≈ 0.65–0.7. Where real and null values differ, real snapshots resemble their predecessor more than randomized snapshots do, which indicates persistent conversation groups rather than reorganisation. This community-level result is distinct from the tie-level overlap of Sec. 5.2.

Sensitivity. The comparison was repeated for three window widths and three resolutions γ. In PS, real snapshots agree less with the static partition than P1 snapshots in all nine settings (differences between -0.03 and -0.23). In CM the difference remains within ± 0.06, and within ± 0.025 except for 30-day windows with $\gamma = 0.5$. The comparison with the previous window is not robust in PS: at 15 and 60 min real snapshots change more than P1 snapshots, whereas at 5 min the order reverses, because a 5-min P1 snapshot is a small random sample of the static graph. Larger γ yields more and smaller communities without changing these signs. Timestamp permutation spreads each node's events over the whole period, so a P1 window contains almost all nodes (240 against 175 active nodes per 15-min PS window); part of the gap to the static partition therefore reflects which nodes are active at a given time, in addition to which nodes interact. The midday decline of the class NMI is unaffected, because it compares real snapshots with external labels.

Verdict on H5. H5 is supported for face-to-face contacts and not supported for online messages. In PS the static partition recovers the classes but conceals the midday mixing, which P1 does not reproduce. In CM the static partition summarises the weekly snapshots as well as it summarises time-shuffled ones, and the snapshots are, if anything, more persistent than random.

5.4. Paths, centrality and spreading (H6, H7)

Temporal reachability (M3a). In PS, time-respecting paths reach 86.6% of ordered pairs, whereas P1 and P2 reach essentially all pairs; the mean latency is 3.39 h against 0.15 h and 0.12 h (Tab. 8). In CM, $R = 0.205$ against 0.326 under P1 and 0.237 under P2, while the mean latency ex-

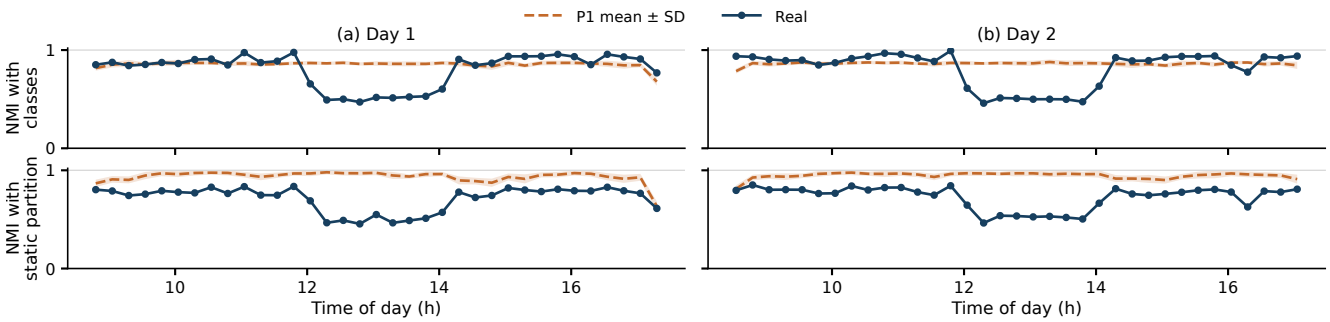


Figure 5. Snapshot communities in PS (15-min windows, $\gamma = 1$): NMI with the classes (top) and with the static partition (bottom). P1: mean $\pm$ SD of 20 samples.

Dataset	Metric	Real	P1	P2
PS	Reachability R	0.866	1.000	1.000
PS	Mean latency (h)	3.390	0.152	0.124
CM	Reachability	0.140	0.223	0.165
CM	Reachability R	0.205	0.326	0.237
CM	Mean latency (h)	948.993	906.718	912.285

Table 8. Multi-start temporal reachability. P1 and P2 columns average 20 networks each; for PS the static reachability is 1, so R equals the reachability.

ceeds the null values by approximately 5%. Both networks therefore have lower reachability than their randomized counterparts. Because late start times have a shorter remaining horizon and PS contacts follow the school timetable, R is not a purely topological quantity; moreover, P2 alters the topology as well as the timing.

Static and temporal centrality (M3b). Static and temporal closeness rankings are positively correlated in both networks (Kendall $\tau = 0.71$ in PS and 0.78 in CM for hop distances; Tab. 9), so the static network identifies which nodes are broadly central. Agreement on the most central nodes is weaker: only half of the PS top 10% and two thirds of the CM top 10% coincide. In CM the agreement is lower than under P1 (0.78 against 0.85), and the τ value is inflated by the 549 users (29%) who never send a message and have zero closeness in both rankings. In PS the result depends on edge weights. Weighted static closeness agrees poorly with temporal closeness ($\tau = 0.39$, top-10% overlap 0.29), whereas after P1 the agreement is high (0.69 ± 0.01). Most contact weight in the school arises from repeated contacts within a class during lessons, when the class is already connected; such contacts hardly shorten time-respecting paths to the rest of the school. Without weights the contrast reverses (real 0.71, P1 0.55), since rapid reach depends on the number of distinct partners, which in PS is largely generated by mixing at breaks.

	Measure	Real	P1
PS	τ, unweighted	0.71	0.55 ± 0.01
	τ, weighted	0.39	0.69 ± 0.01
	Top-10% overlap, unweighted	0.50	0.51 ± 0.07
	Top-10% overlap, weighted	0.29	0.60 ± 0.06
CM	τ, unweighted	0.78	0.85 ± 0.00
	τ, weighted	0.74	0.86 ± 0.00
	Top-10% overlap, unweighted	0.65	0.74 ± 0.01
	Top-10% overlap, weighted	0.57	0.72 ± 0.02

Table 9. Agreement of static and temporal closeness rankings (Kendall τ_b and overlap of the top 10% of nodes). P1: mean $\pm$ SD over 20 samples; the static ranking is unaffected by P1.

Event-driven SIR (M3c). Table 10 reports the SIR outcomes for the four parameter settings of each network. In PS the final infected fraction is smaller than under both nulls in all four settings (0.629 against 0.776 under P1 at the primary setting), and the peak ratio increases with the transmission probability: 1.000 at $\beta = 0.02$, 1.160 at 0.05 and 1.311 at 0.10 for a four-hour infectious duration (ratios relative to P1). The qualifying populations differ (major outbreaks occur for 0.860 of the real seeds against 0.908 under P1), so the peak comparisons are not paired-seed estimates. In CM the peak occurs earlier than under P1 in all full-period settings (ratios 0.72–0.94) and is approximately unchanged relative to P2 (0.98–1.01). The choice of timing metric also matters: at $\beta = 0.5$ and 14 days the peak ratio to P1 is 0.924, whereas the time to half of each trajectory's final size has a ratio of 1.061. These opposing signs preclude a general statement that spreading is faster or slower in the real network.

Verdict on H6 and H7. The reachability component of H6 is supported in both networks under the multi-start protocol, and the ranking component is partially supported: the rankings are correlated, but the top nodes differ and weighting by contact frequency makes the static PS rank-

Dataset	(β, D)	Final infected fraction			Major outbreak probability			Peak ratio	
		Real	P1	P2	Real	P1	P2	vs P1	vs P2
PS	(0.05, 4 h)	0.629	0.776	0.823	0.860	0.908	0.920	1.160	1.214
PS	(0.02, 4 h)	0.373	0.420	0.559	0.700	0.754	0.778	1.000	1.088
PS	(0.10, 4 h)	0.734	0.893	0.918	0.910	0.960	0.973	1.311	1.340
PS	(0.05, 1 h)	0.373	0.634	0.674	0.745	0.844	0.852	1.021	1.265
CM	(0.20, 336 h)	0.256	0.250	0.276	0.825	0.791	0.799	0.888	0.995
CM	(0.05, 336 h)	0.026	0.013	0.025	0.215	0.082	0.192	0.934	0.991
CM	(0.50, 336 h)	0.460	0.506	0.530	0.975	0.954	0.965	0.924	0.982
CM	(0.20, 72 h)	0.074	0.043	0.066	0.555	0.342	0.428	0.720	1.013

Table 10. SIR outcomes for four (β, D) settings per dataset (200 seeds per network, 20 nulls per family). The first setting of each dataset is primary. The peak ratio is restricted to major outbreaks, whose populations can differ between real and null networks.

ing misleading. H7 is conditionally supported in PS at its primary setting, with a negligible peak effect at low transmission probability and a reversal for the one-hour setting in the first half. The smaller final size is a result separate from the delayed peak. CM does not support the hypothesized slowdown: full-period peaks occur earlier than under P1 and close to P2, and the peak timing can differ in sign from the half-size timing. The verdict thus depends on the null model, the outcome metric, the parameters and the observation interval, and the ratios, which condition on network-specific major-outbreak subsets, do not establish a population-wide causal slowing effect.

5.5. Robustness and sensitivity

Citation network. Table 11 reports time splits, a 2015 cutoff and corpus-selection checks. The positive degree preference and the superlinear full-period growth persist in all variants, whereas densification is substantially weaker in the later period. Degree offsets of 0.5 and 2 yield $\alpha = 0.604$ and 0.858, so the numerical exponent depends on this convention. Including the 5,486 same-year citations yields $a = 1.324$. The effective diameter equals 6 hops with 64, 128 or 256 sources. The mean null exponents over the first 10 and all 20 samples are 0.00055 and 0.00059 for P3a and -0.65320 and -0.65118 for P3b. Lognormal ageing is preferred in all publication cohorts and in both observation halves. These checks indicate qualitative stability but do not remove coverage bias, right censoring or unobserved fitness. Finally, in-sample fit quality of the WSB model must be distinguished from predictive skill: the full-history sample comprises 500 of 1,019 eligible papers (median scaled-collapse RMSE 0.022; 10.2% of fits at a parameter bound), whereas only the held-out comparison of Tab. 7 tests prediction.

Contact and messaging networks. Table 12 reports the elapsed-time halves and the enlarged null ensemble. Increasing the ensemble from 20 to 60 networks changes the primary results little, whereas the observation interval is more consequential: the CM H3 contrast changes from $+0.027$ to -0.039 and -0.022 in the halves, and the PS H2 excess from 0.016 to 0.116 and 0.120, whereas the H1 sensitivities on the common window grid change little. The primary H7 ratios keep their direction under the 20/60 comparison (PS 1.160 to 1.161, CM 0.888 to 0.884) and in both halves, but not uniformly across parameter settings: at PS $\beta = 0.02$ the ratio moves from 1.0005 to 0.9992, and the PS one-hour setting peaks earlier than under P1 in the first half (ratio 0.874). In the second half of CM, two settings produce no major outbreak in the real network, so their ratios are undefined; the primary setting remains defined (major-outbreak probability 0.360). The PS final size is 0.629 over the full interval and 0.821 and 0.788 in the halves; because seeds are redrawn from the first 70% of each interval, the seed populations and residual horizons change, and these comparisons do not separate schedule effects from censoring. Splitting also alters node populations and the time available for paths and outbreaks, so agreement in sign is more informative than equality of effect sizes.

Variant	α	a
1990–2020 (primary)	0.714	1.343
1990–2004	0.684	1.455
2005–2020	0.721	1.065
Cutoff 2015	0.722	1.427
Core venues	0.731	1.410
Same-year edges included	–	1.324

Table 11. CIT robustness of the attachment exponent α and the densification exponent a. The core venues contain 14,235 nodes and 47,009 primary edges.

5.6. Cross-network summary

Table 13 reports the time-effect measure of each hypothesis together with the verdict reached in the preceding subsections. The measures differ in units and interpretation, so the table does not define a common sensitivity score for a

	Measure	Full	First	Second	60 nulls
PS	H1 density	2.503	2.551	2.518	–
CM	H1 density	5.411	5.403	5.190	–
PS	H1 clustering	1.394	1.470	1.487	–
PS	H2 ΔB	0.016	0.116	0.120	0.016
CM	H2 ΔB	0.336	0.381	0.313	0.336
PS	H3 ΔJ	-0.147	-0.192	-0.184	-0.147
CM	H3 ΔJ	0.027	-0.039	-0.022	0.027
PS	H5 NMI	0.816	0.812	0.818	0.816
CM	H5 NMI	0.519	0.374	0.663	0.519
PS	H6 R	0.866	0.815	0.765	0.866
CM	H6 R	0.205	0.314	0.177	0.205
PS	H7 ratio vs P1	1.160	1.081	1.095	1.161
CM	H7 ratio vs P1	0.888	0.856	0.865	0.884
PS	H7 ratio vs P2	1.214	1.142	1.160	1.212
CM	H7 ratio vs P2	0.995	0.961	0.877	0.999
CIT	H4 α	0.714	0.684	0.721	0.714

Table 12. Elapsed-time halves (PS, CM) and analysis-year halves (CIT). H1 uses the widths with complete windows in both halves (PS without 24 h) and has no reference model. H2 and H3 are real minus P1; H5 and H6 show the observed adjacent-snapshot NMI and R; H7 shows conditional peak ratios. The 60-null ensemble contains the 20 primary networks.

network type. Temporal information is essential for every hypothesis in at least one network, but the direction and robustness of the effect vary with the system and, for H2, H3 and H7, with the event definition, window width, observation interval or reference model.

6. Discussion

Information retained by the static aggregate. The static aggregate preserves which pairs interact and, for the citation network, the degree heterogeneity, clustering and connected components of the final graph. Because static and temporal closeness rankings are positively correlated in both PS and CM, it also identifies which nodes are broadly central. For questions of this kind a static representation is adequate.

Information that requires time. All other conclusions depend on temporal information, and they do so at increasing levels of detail. The choice of window alone changes most structural metrics by more than their mean value (H1). Activity patterns, community dynamics and network function require the full event sequence. In CM, burstiness clearly exceeds the timestamp-shuffled baseline, whereas in PS the heavy-tailed intervals are largely produced by the school schedule, which the shuffled baseline retains. Link turnover cannot be represented by a single graph, the midday mixing in PS is invisible in the static partition, and time-respecting paths reach fewer nodes than the static graph suggests. The most central nodes also differ between static and temporal rankings; weighting the static graph by contact frequency makes the PS ranking misleading, because repeated within-class contacts do not shorten time-respecting paths to the rest of the school. For citations, attachment, ageing and densification can be measured only from the historical sequence of degrees, publication ages and $(N(t), E(t))$ pairs.

Differences between network types. The three networks differ in generating mechanism, time scale and edge semantics. A contact ends, a message is instantaneous and a citation persists. PS interactions are constrained by the school timetable and by physical proximity, which produces fast turnover, a strong class structure interrupted at midday and, at the primary setting, epidemic peaks later than under both reference models. CM records voluntary communication over a much longer period; its stronger excess burstiness is consistent with conversational exchanges, while its sparse weekly snapshots show neither community reorganisation nor a spreading slowdown. CIT has persistent edges and is acyclic by construction, so time enters through growth, ageing and densification rather than through paths or turnover. The relevance of time is therefore specific to the hypothesis rather than to the network type, and direct comparisons between networks remain conditional on their event definitions and observation periods.

Dependence on protocol choices. Several conclusions depend more on the observation interval and on the definition of events and windows than on the size of the null ensemble (Sec. 5.5). The PS burstiness excess changes with the observation interval and the event definition, and the sign of the CM persistence contrast changes with the observation interval and the window width (Tab. 5). Timestamp permutation redistributes events over the selected interval, so its baseline varies with the duration, activity and window composition of that interval. The H7 verdict further depends on the reference model, the parameter setting and the timing metric, and it conditions on network-specific subsets of major outbreaks. Quantities that are invariant under P1, such as the CM repeated-tie fraction, are descriptive, and their near-zero null dispersion is not evidence of a temporal effect. In the citation network, the attachment exponent depends on the degree offset and the densification exponent on the period, whereas the signs of both and the diameter result are stable. These checks establish qualitative stability but do not remove coverage and censoring effects.

Description, mechanism and prediction. The citation results separate claims that are often combined. The cumulative edge count grows superlinearly, but the sampled effective diameter does not shrink. The WSB model fits selected citation histories closely in sample, yet its forecast

ID	Measure	PS	CM	CIT	Verdict
H1	Density sensitivity	3.262	5.411	1.798	Supported in all three networks
H2	ΔB vs P1	0.016	0.336	–	Burstiness supported in CM; small and definition-dependent in PS; memory not supported
H3	ΔJ vs P1	-0.147	0.027	–	Turnover supported in PS; CM persistence depends on width and interval
H4	$\Delta\alpha$ vs P3a	–	–	0.713	Attachment, ageing and densification supported; no diameter shrinkage or forecast advantage
H5	Δ adjacent NMI vs P1	-0.096	0.098	–	Supported in PS; not supported in CM
H6	ΔR vs P1	-0.134	-0.120	–	Reachability restricted in both; rankings correlated but top nodes differ
	ΔR vs P2	-0.134	-0.031	–	
H7	Peak ratio vs P1	1.160	0.888	–	Conditionally supported in PS; not supported in CM
	Peak ratio vs P2	1.214	0.995	–	

Table 13. Time effects and verdicts. H1 is descriptive and uses different window grids per network (PS 5 min–24 h, CM 1 h–30 d, CIT 1–10 years), so its values cannot be ranked. H5 uses 15-min (PS) and 7-day (CM) windows. H7 reports the primary SIR setting and conditions on major outbreaks. A dash denotes inapplicability and not a zero effect.

advantage over linear extrapolation is not established. The zero annual clustering is a consequence of the strict-year snapshot construction. Representation, in-sample description and held-out prediction therefore require separate assessment.

Limitations. The scope of the conclusions is limited by the data. Each network type is represented by a single dataset, the contact data cover only two days and therefore introduce boundary effects in tie lifetimes, and each system contains several time scales [12], so a window suited to one process may obscure another. The reference models impose further limits. Timestamp permutation preserves the global timestamp distribution and therefore does not remove all nonstationarity, endpoint swapping is not guaranteed to sample uniformly from the admissible networks, and 20 realizations per model provide only a coarse tail resolution. Pooled burstiness and memory may also reflect heterogeneous edge activity, and reachability includes finite-horizon effects. The citation network is a degenerate temporal network, acyclic and without deletion, which excludes several methods. Its conclusions are additionally constrained by within-venue sampling, annual timestamp ties and changing reference coverage; the bootstrap intervals are descriptive because cumulative observations are dependent, and forecast selection excludes weakly cited papers, with 38.6% of short-history fits reaching parameter bounds.

7. Conclusion

This study tested seven hypotheses from the temporal-network literature on a contact, a messaging and a citation network within a common framework. The importance of temporal information depends on the question asked. In the citation network, the final aggregate describes degree heterogeneity and connectivity, whereas preferential attachment, ageing and the growth law require historical degrees, publication ages or cumulative snapshots. Within the selected AMiner corpus, sublinear degree preference and age-dependent citation choice coexist with superlinear edge growth, but the sampled effective diameter does not shrink. The WSB model describes citation histories compactly without an established forecast advantage over linear extrapolation, so descriptive fit, temporal mechanism and predictive value must be assessed separately.

In the contact and messaging networks, structural metrics depend strongly on the window width, and time ordering restricts reachability over a finite horizon. Burstiness exceeds the timestamp-shuffled baseline robustly in CM, whereas in PS the excess is small and depends on the observation interval and the event definition; memory is weaker than under the baseline in both networks. The persistence of CM ties relative to the baseline reverses within both observation halves and at 14-day windows. Static communities conceal the midday mixing in PS, while CM shows no comparable reorganisation. Conditional on major outbreaks, epidemic peaks in PS occur later than under both reference models at the primary setting, whereas CM does not support a general slowdown. These hypothesis-specific effects cannot be combined into a single ranking of network types without further assumptions. Several datasets per network type, broader corpus coverage, finer-resolution citation dates and predictive tests under alternative censoring assumptions would strengthen the conclusions.

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